Executive Summary

TikTok Project: Predicting Verified vs. Unverified Accounts
Logistic Regression Modeling & Evaluation Metrics Results

Project Overview:

In this phase of the project, exploratory data analysis (EDA) and logistic regression modeling were applied to analyze TikTok video characteristics and their relationship with **verified_status**. This analysis aims to reveal how specific video features relate to verified users, informing the classification pipeline.

Key Insights:

• **Model Performance:** After scaling numerical features (StandardScaler) and adjusting for class balance (class_weight="balanced"), the logistic regression model achieved a strong **recall of 84%** for verified accounts, effectively identifying verified users across video metrics.

• **Video Duration:** Video duration (video_duration_sec) shows a significant positive association with verified status. For every additional second in video duration, the log-odds of an account being verified increase by 0.009.

• **Impact of Engagement Metrics:** Other features (view count, share count, download count, comment count) yield smaller coefficients, indicating secondary association compared to video length and claim status.

Details & Methodology:

• **Methodology:** Exploratory Data Analysis (EDA), feature scaling (StandardScaler), feature encoding (One-Hot-Encoding), class weight balancing, and logistic regression modeling using Python (scikit-learn).

• **Evaluation Metrics:** Precision, Recall, F1-Score, Accuracy, and Confusion Matrix analysis.

• **Independent Variables:** video_duration_sec, video_view_count, video_share_count, video_download_count, video_comment_count, claim_status, and author_ban_status.

Next Steps:

1. **Model Integration:** Incorporate video duration and engagement features into tree-based ensemble models (Random Forest / XGBoost) to further improve precision and recall.

2. **Algorithmic Bias Audit:** Conduct regular audits on models relying on duration metrics to avoid unintentional bias against short-form creators.